

NETWORKS AND COMPLEXITY

Exercise Sheet 21: The Vibe of a City

*This is an exercise sheet from the forthcoming book *Networks and Complexity*.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 21.1: Laplacian construction [Lvl. 1]

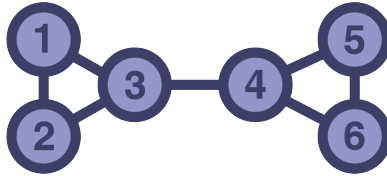
Construct the matrices $\mathbf{A}$, $\mathbf{L}$, $\mathbf{L}_{\text{rw}}$, $\mathbf{L}_{\text{sym}}$ for a cycle of four nodes.

Ex 21.2: Random derivation [Lvl. 2]

Consider a random walk on a network described by an adjacency matrix $\mathbf{A}$. A walker that in a node i leaves at a random point in time such that the departure rate equals the node degree k_i . When leaving the walker picks a link connecting to i at random and moves instantaneously to the link's other endpoint j . Derive a differential equation that governs the dynamics of x_i , the probability that the walker is in node i .

Ex 21.3: Barbell graph [Lvl. 3]

Consider a barbell graph



Construct the Laplacian $\mathbf{L}$, compute its eigenvalues and eigenvectors. Identify the algebraic connectivity λ_2 , compare it to the Cheeger's constant g and identify the cut implied by the Fiedler vector $\mathbf{v}_2$.

Ex 21.4: Spectral methods in the wild [Lvl. 3]

A supermarket chain approaches you. Their goal is to identify different groups in their customer base. They have a large dataset containing 1 million checkout receipts. Each receipt contains information which products the customer has bought in the purchase. Describe in bullet points how you would approach this question.

Ex 21.5: A different random walk [Lvl. 3]

Consider a different continuous-time random walk on a network: With rate $r = 1$ the walker decides to move. Then it picks one of the links that are available from its current node and instantaneously moves to that link's destination.

- Consider this walk on the 3-chain, (1)-(2)-(3). Define x_i as the probability that the walker is currently in node i . Derive the differential equations that govern these variables and show that they can be written as

$$\dot{\mathbf{x}} = -\mathbf{L}_{\text{rw}}^T \mathbf{x},$$

where $\mathbf{x} = (x_1, x_2, x_3)^T$.

- Now show the same for a general network where the network topology is given by an adjacency matrix $\mathbf{A}$.

Ex 21.6: Laplacian as second derivative [Lvl. 3]

Consider diffusion in a one-dimensional continuous space that is the interval $[0, 5]$. In continuous space diffusion is described by the Laplace equation

$$\dot{x} = \Delta x = \frac{\partial^2}{\partial l^2} x$$

- a) Discretize the interval into five bins, numbered 1 to 5. Each bin i has an associated variable x_i that counts the number of walkers between $l = i - 1$ and $l = i$. Show that the dynamics in bin 3 is approximated by

$$\dot{x}_3 = -2x_3 + x_2 + x_4.$$

- b) Compare this result with the random walk on a network that consists of a chain of five nodes, numbered again from 1 to 5. What is the differential equation for x_3 , the number of walkers in node 3 in this model?

Ex 21.7: Laplacian properties [Lvl. 4]

For undirected graphs the Laplacian $\mathbf{L}$ has an interesting property: If we multiply it from both sides with a vector $\mathbf{x}$ we find

$$\mathbf{x}^T \mathbf{L} \mathbf{x} = \sum_{i,j>i} A_{ij} (x_i - x_j)^2.$$

which sums the quadratic differences in x over all links, providing a measure of the smoothness of $\mathbf{x}$.

- a) Show that this equation is correct.
 b) Also find interpretations for the following expressions: $(\mathbf{L}\mathbf{x})_i$, $(\mathbf{L}_{\mathbf{rw}}\mathbf{x})_i$, $\mathbf{x}^T \mathbf{Q} \mathbf{x}$, $|\mathbf{Q}|$, $\text{Tr}(\mathbf{L})$, $\text{Tr}(\mathbf{L}_{\mathbf{rw}})$, where $\mathbf{Q} = \mathbf{K} + \mathbf{A}$ is the *signless Laplacian*.
 c) What would $|\mathbf{Q}| = 0$ imply?

Ex 21.8: Second eigenvalue [Lvl. 4]

In a weighted network the weighted adjacency is given by

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}$$

Compute the corresponding Laplacian. Then, find an iterative method to compute the Fiedler vector $\mathbf{v}_2$.

Ex 21.9: Sorting numbers [Lvl. 3]

We say numbers are similar if they have a divider in common that is greater than 1. So 2 and 3 are not similar, but 2 and 10 are similar, because they are both divisible by 2. Construct a network where the nodes represent the numbers from 1 to 10 and links indicate that the connected numbers are similar. Then construct the Laplacian and find eigenvectors for the four eigenvalues closest to zero. Do the cuts implied by these eigenvectors make sense?

Ex 21.10: Leaky conveyor [Lvl. 4]

Consider a random walker on a network described by a weighted, directed adjacency matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix}$$

the element A_{ij} describes the rate at which the walker that is in node j jumps to node i . Determine the probability that the walker is in node 1 in the stationary state. [The probabilities become stationary, not the walker.]

Ex 21.11: Discrete-time diffusion [Lvl. 4]

Consider a discrete-time random walk on a network where in each timestep the walker picks one of the neighbors of its current node and moves there. Let $x_i(t)$ be the probability that the walker is in node i in t .

- a) Study this process first on the 3-chain, (1)-(2)-(3). Write a discrete time map that captures the changes of the probabilities, then solve it for a walker starting in node 1 at $t = 0$.
- b) Now find the general solution for this discrete time random walk on an arbitrary network described by an adjacency matrix $\mathbf{A}$.

Ex 21.12: Path graph [Lvl. 4]

Consider a long chain of nodes that are numbered consecutively from 1 to N . This is a so-called path graph.

- a) Consider the eigenvector equation for the adjacency matrix in node j in the middle of the chain and use the Ansatz $v_j = ab^j$, and find possible values for b . Did you notice something odd? Compute $|b|$.
- b) Let's define θ such that $\lambda = 2\cos(\theta)$. Explain why this is an intuitive substitution. Then substitute it into b . Then an almost magical simplification is possible.
- c) The adjacency matrix is a real symmetric matrix and hence should have real eigenvectors, but neither of our solutions for b gives us such a vector. However, show that a linear combination of the solutions is also a solution.
- d) At the beginning and end of the chain, the eigenvector condition is slightly different. For example in node 1: $v_2 = \lambda v_1$. However, we can write it in the familiar form $v_0 + v_2 = \lambda v_1$, if we demand $v_0 = 0$. Hence introduce *ghost nodes*, numbered 0 and $N + 1$ and demand that $v_0 = v_{N+1} = 0$. Use this boundary condition to find all eigenvalues and eigenvectors.